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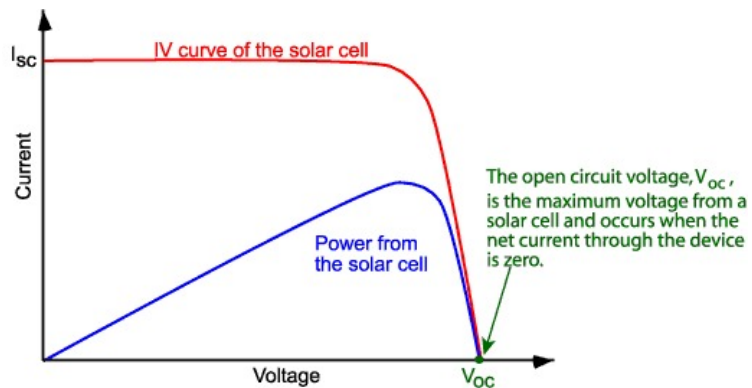
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Open-Circuit Voltage

The open-circuit voltage, V_{OC} , is the maximum voltage available from a solar cell, and this occurs at zero current. The open-circuit voltage corresponds to the amount of forward bias on the solar cell due to the bias of the solar cell junction with the light-generated current. The open-circuit voltage is shown on the IV curve below.



IV curve of a solar cell showing the open-circuit voltage.

An equation for V_{OC} is found by setting the net current equal to zero in the solar cell equation to give:

$$V_{OC} = \frac{nkT}{q} \ln \left(\frac{I_L}{I_0} + 1 \right)$$

A casual inspection of the above equation might indicate that V_{OC} goes up linearly with temperature. However, this is not the case as I_0 increases rapidly with temperature primarily due to changes in the intrinsic carrier concentration n_i . The effect of temperature is complicated and varies with cell technology. See the page "Effect of Temperature " for more details

V_{OC} decreases with temperature. If temperature changes, I_0 also changes.

Open Circuit Voltage Calculator 1

Dark Saturation Current, I_0 = A

Light Generated Current, I_L = A

Ideality Factor, n =

Temperature, T = K

Open Circuit Voltage, V_{OC} = V

The above equation shows that V_{OC} depends on the saturation current of the solar cell and the light-generated current. While I_{sc} typically has a small variation, the key effect is the saturation current, since this may vary by orders of magnitude. The saturation current, I_0 depends on recombination in the solar cell. Open-circuit voltage is then a measure of the amount of recombination in the device. Silicon solar cells on high quality single crystalline material have

open-circuit voltages of up to 764 mV under one sun and AM1.5 conditions¹, while commercial silicon devices typically have open-circuit voltages around 690 mV.

The V_{OC} can also be determined from the carrier concentration²:

$$V_{OC} = \frac{kT}{q} \ln \left[\frac{(N_A + \Delta n)\Delta n}{n_i^2} \right]$$

where kT/q is the thermal voltage, N_A is the doping concentration, Δn is the excess carrier concentration and n_i is the intrinsic carrier concentration. The determination of V_{OC} from the carrier concentration is also termed Implied V_{OC} .

Open Circuit Voltage Calculator 2

Doping Concentration, $N_A = 1.5e16$ cm⁻³
 Excess Carrier Concentration, $\Delta n = 1e15$ cm⁻³
 Temperature, $T = 298$ K
 Intrinsic Carrier Concentration, $n_i = 8.6e9$ cm⁻³
 Open Circuit Voltage, $V_{OC} = 0.667$ V

Voc as a Function of Bandgap, E_G

Where the short-circuit current (I_{SC}) decreases with increasing bandgap, the open-circuit voltage increases as the band gap increases. In an ideal device the V_{OC} is limited by radiative recombination and the analysis uses the principle of detailed balance to determine the minimum possible value for J_0 .

The minimum value of the diode saturation current is given by³:

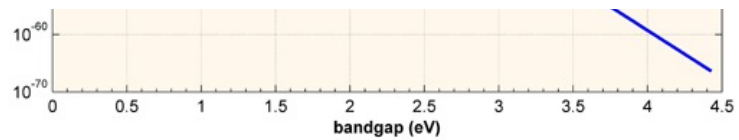
$$J_0 = \frac{q}{k} \frac{15\sigma}{\pi^4} T^3 \int_u^\infty \frac{x^2}{e^x - 1} dx$$

where q is the electronic charge, σ is the Stefan–Boltzmann constant, k is Boltzmann constant, T is the temperature and

$$u = \frac{E_G}{kT}$$

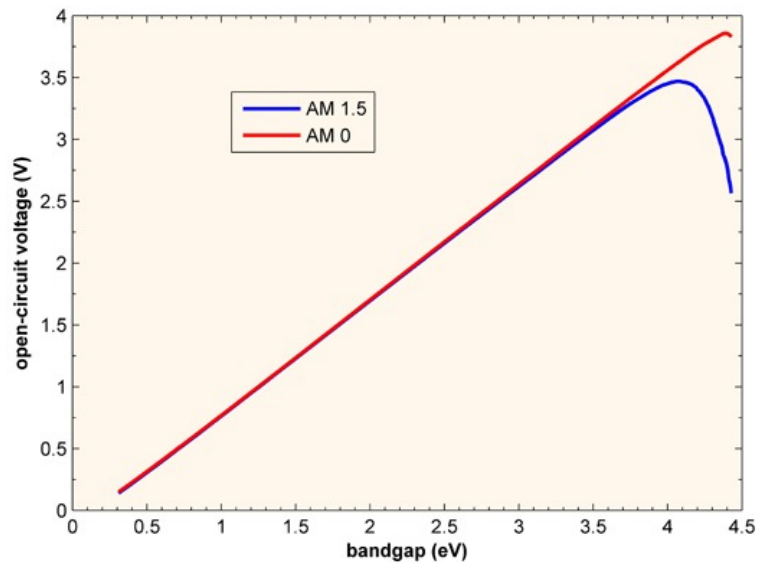
Evaluating the integral in the above equation is quite complex. The graph below uses the method outlined in⁴





Diode saturation current as a function of band gap. The values are determined from detailed balance and place a limit on the open circuit voltage of a solar cell.

The J_0 calculated above can be directly plugged into the standard solar cell equation given at the top of the page to determine the V_{OC} so long as the voltage is less than the band gap, as is the case under one sun illumination.



V_{OC} as function of bandgap for a cell with AM 0 and AM 1.5. The V_{OC} increases with bandgap as the recombination current falls. There is drop off in V_{OC} at very high band gaps due to the very low I_{SC} .

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